01 What I built and why

Five Python stages: classify -> page-cited retrieval from three locked PDFs -> admit/plan -> stream via Groq or stand-in -> verify/refuse. Deliberate trade-off: refuse unsupported claims instead of generating fast uncited answers.

02 The real service and my stand-in

Groq V3: 18 paced HTTP200 streaming calls, content TTFT p50 322.9 ms; V1: 16 successes then one HTTP429 at the token/min boundary (retry-after 5s). Headers showed 8,000 TPM and 1,000 RPD. V2's 18 HTTP401 failures came from an escaped secret, and are NOT successful real probes. Stand-in fits V1 median TTFT, not unmeasured parallel slowdowns.

TABLE 1 / Provider limits — measurement versus published documentation

Limit	Value found	How discovered	Confidence
Requests/min	30 (docs)	Public free plan; not challenged	medium
Tokens/min	8,000	x-ratelimit-limit-tokens, HTTP200/429	high
Requests/day	1,000	x-ratelimit-limit-requests is RPD	high
Tokens/day	200,000 (docs)	No account TPD header measured	medium
In flight at once	not measured	Only sequential hosted calls	low
Limit response	HTTP 429; 5 sec	V1 17th request retry-after=5	high
Real recovery	not measured	5sec suggested, not tested	low

03 Results

Milliseconds p50/p95/p99. First token = first visible model content, only for requests emitting content (not a fully verified user reply); NM = not measured. Calibrated stand-in load shapes remain modelled, not observed Groq concurrency.

TABLE 2a / First token and completion time by traffic shape

Scenario	Source	N	Duration s	Offer/min	TTFT p50/p95/p99 ms	Total p50/p95/p99 ms
Real Groq V3	Groq	18	91.50	11.8	322.9/454.7/496.4	348.5/475.5/528.3
Steady	stand-in	12	3.85	186.8	1079.9/1270.5/1347.5	953.5/1304.4/1394.4
Burst	stand-in	22	0.73	1797.8	270.2/677.5/694.9	250.9/649.7/716.5
Sustained	stand-in	28	1.94	867.6	614.1/1287.7/1351.7	251.1/1149.1/1362.8
RPM halved	stand-in	16	2.02	475.2	698.9/1303.9/1322.6	251.5/1321.2/1373.7
RPD exhausted	stand-in	10	0.54	1106.2	NM/NM/NM	0.4/0.5/0.5

TABLE 2b / Delivery outcomes and costs from the same per-request file

Scenario	Success	Rate-limited	Timeout/error	Retries/req	Peak queue	Tokens/answer	USD/1k*
Real Groq V3	100.0%	0.0%	0.0%	0.00	0	530.1	0.052
Steady	83.3%	0.0%	16.7%	0.00	0	96.9	0.000
Burst	27.3%	0.0%	72.7%	0.00	13	18.9	0.000
Sustained	32.1%	0.0%	67.9%	0.00	4	29.5	0.000
RPM halved	43.8%	0.0%	56.2%	0.00	1	45.9	0.000
RPD exhausted	0.0%	100.0%	0.0%	0.00	0	0.0	0.000

* Counterfactual retail pricing; no free-tier bill

TABLE 3 / Three full-PDF retrieval settings, 18 questions each; strict string/citation oracle

Chunk / K / rerank	N	Correct+cited	Wrong	Correct refuse	Token proxy/answer	TTFT p50/p95 ms	Total p50/p95 ms
220 chars / top1 / no	18	33.3%	55.6%	0.0%	104.6	106.8/163.4	266.4/359.6
500 chars / top3 / yes	18	33.3%	61.1%	0.0%	379.1	113.9/159.9	307.5/396.9
1000 chars / top6 / yes	18	33.3%	61.1%	0.0%	1394.9	109.8/153.4	288.5/415.8

Full-PDF stand-in: 18/54 strict correct, 32/54 wrong, 4/54 wrongly refused. V3 real Groq: 6/18 strict correct. Provisional AI semantic audit suggests 12/18 correct+cited, 3 justified refusals, 2 unnecessary refusals, 1 wrong; NOT independently human-verified.

04 Where the time goes

Slow Groq V3 request real-002: total 541.5ms; first content 506.9ms. Local retrieval/plan 8.1ms, HTTP acceptance/server interval 446.6ms, accepted-to-content 52.1ms, post-content stream+verify 34.6ms, 0 retries. Groq's internal queuing is not separately measurable.

05 When a limit is hit

At TPM, shrink evidence and check whether Retry-After fits the waiting person's deadline; otherwise reject explicitly. At RPD exhaustion, stop hosted calls and never silently downgrade into an uncited guess. V1 measured a five-second retry-after, but not actual recovery.

Retrieval: quality versus cost

Three tested settings: 220 chars/top-1, 500 chars/top-3, 1,000 chars/top-6. Each strict full-PDF stand-in run scored only 6/18, though wide context cost far more tokens. Prefer the balanced K=3 for budget, but this correctness is not adequate for unattended deployment. Near budget: smaller K or refusal.

How this library misleads a naive retriever

Attention heads appear in multiple papers; latency passages mix Moshi's 160ms theoretical value with other delay values; PDF extraction can break mathematical formula text such as sqrt(dk). Paper routing and version/page citations mitigate, but do not solve, these errors.

06 Retries

One pre-first-token transient 503 retry recovered an answer but took longer. Never restart a partially streamed reply. Retrying a measured 429 immediately would worsen latency, since the returned retry-after was 5 seconds.

07 What the person sees

At 3pm when the day quota is exhausted: "I cannot provide a verified answer now because the daily hosted-model quota is exhausted. Please try after reset." Do not fabricate a paper citation.

08 What I am unsure about, left out, learned

Least certain: the stand-in's concurrency slowdown; all Groq calls were serial. Wrong assumption breaks p95 overload. Deliberately omitted public multi-tenant auth and an embedding reranker. Lesson: a 200 status need not mean a correct answer; minute tokens can run out long before daily requests.

Your feedback on Vaani

The Vaani developer guide's distinction between dispatch and completion stands out. Latency definitions (interrupt vs full response) are difficult to compare; a common glossary showing p50/p95, load, sample size and measurement boundaries would help. Applicant should personally review this opinion.

How I worked with the assistant

The original user supplied the brief and GitHub repository, then configured a Groq Secret and sent the run link. The assistant wrote code, results and this report. The first optimistic fixture metric failed on real PDFs; an assistant-introduced V2 secret bug caused HTTP401 until detected and fixed. Applicant must include unedited conversation history and verify personal decisions before submission.

Evidence and limitations

447 per-request rows retain V1 (including true 429), V2 (invalid 401s), V3 (18 real HTTP200s), earlier uncalibrated and newly calibrated stand-in runs. Cost is hypothetical list price, not charge. Groq recovery, true in-flight limit and external human semantic grading remain unmeasured.